With its scalability, security, and efficiency, netbooting remains an excellent choice for organisations looking to optimise system management and integrate AI-driven technologies seamlessly into large-scale deployments. Whether for LLM-based interactive environments, AI inference clusters, or high-performance computing, netbooting continues to evolve as a foundational technology for next-generation IT infrastructure.

To facilitate such a working system there are both wired and wireless ways. While wired interfaces require just an Ethernet connection of the host and server, the wireless mechanism requires setting up a DHCP server or modification of the existing DHCP server of the network. Let us explore the wireless netbooting of the LLM model Gemini AI using the steps mentioned below.

First, we need to set up the server with a TFTP server and a DHCP server using the software called SERVA. After extracting the software, we can configure the DHCP server to assign an IP address to the host system using *Serva64.exe* for 64-bit systems, as shown in Figure 1.

This setup can function as a DHCP server with a static IP assignment or, as in this case, as a proxy DHCP server to avoid interfering with the existing DHCP server on the LAN router. Additionally, the BINL protocol, used in Windows Deployment Services (WDS) for PXE booting, facilitates the retrieval of boot images and configurations.

Next, the TFTP server must be configured by specifying the directory where *Serva64.exe* is located, ensuring proper file transfer and deployment, as shown in Figure 2.

Ensure that the block size is adjusted according to the host system's available storage capacity. After configuring both servers, restarting Serva64 will automatically generate the necessary configuration folders within the specified directory.

Next, extract the ISO and insert the files inside the *NWA_PXE* folder and include the *ServaAsset.inf* file alongside the ISO. In this instance, Ubuntu Live Server is being used.

To integrate an AI-based LLM onto the ISO, it is essential to understand the Ubuntu Server boot process. Initially, GRUB or another bootloader is loaded, which then loads the kernel (vmlinuz) and initrd (initial RAM disk). Subsequently, the *vmlinuz* and *initrd* files located inside the *Casper* folder are executed. This process eventually mounts the SquashFS, which contains the Ubuntu environment within the same directory.

To enable custom LLM functionality, a custom Bash script must be incorporated into the SquashFS. This step is crucial as system updates and essential packages must be preloaded into the OS for interacting with the Gemini API and utilising networking tools such as ping or curl.

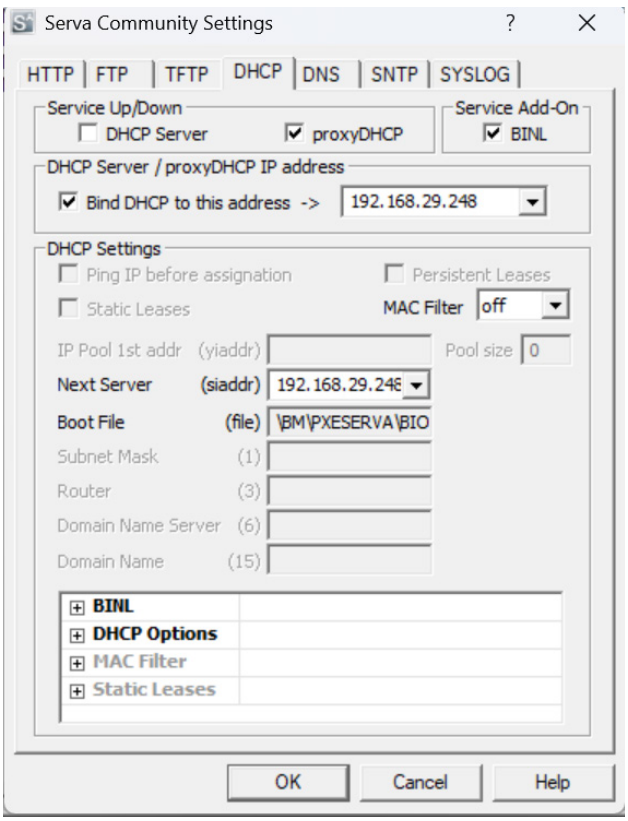

Figure 1: DHCP server configuration

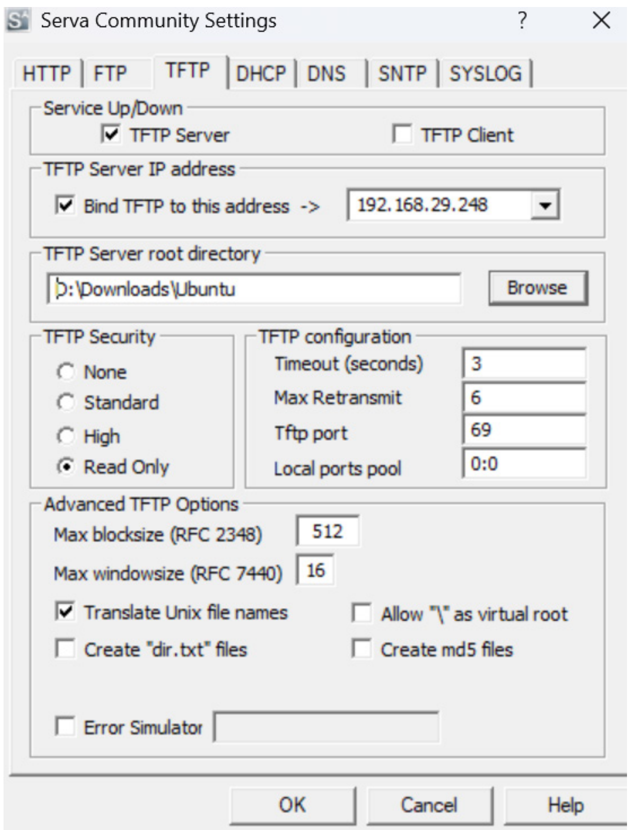

Figure 2: TFTP server configuration